

data manipulation

1.clear workspace

```
rm=(list=ls())
```

2.loading data from package

```
data("mtcars") ?mtcars
```

```
cars<-mtcars
```

3.structure of the data

```
str(cars)
```

4.checking for missing values

```
any(is.na(cars)) sum(is.na(cars))
```

5.Fetching top 6 rows

```
head(cars)
```

fetching last 6 rows

```
tail(cars)
```

6.Viewing data

```
View(cars)
```

tibble_df(cars)

```
as_tibble(cars) glimpse(cars)
```

7.filtering based on single condition

```
filter(cars,mpg>30)
```

8.filtering based on multiple condition

```
filter(cars,mpg>25 & hp>90)
```

9.remove duplicate rows

```
distinct(cars)
```

10.Randomly selecting the fraction of rows

```
?sample_frac sample_frac(cars,0.8)
```

randomly select no.of .rows

```
sample_n(cars,5) slice_sample(cars,n=5)
```

11.select rows by position

```
slice(cars,2:4) slice(cars,16:20)
```

12.selecting single column

```
select(cars,mpg)
```

13.selecting multiple column

```
select(cars,mpg,gear,cyl) select(cars,c("mpg","cyl","gear"))
```

select all columns between a range of column(inclusive)

```
select(cars,hp:am)
```

14.combing filter and select-using pipe operator

```
cars %>% filter(mpg>18)%>% dplyr::select(mpg,am)
```

15.selecting columns starting with 'd'

```
select(cars,starts_with('d'))
```

selecting columns ending with 't'

```
select(cars,ends_with('t'))
```

selecting columns containing 'g'

```
select(cars,contains('g'))
```

selecting columns matching regular expression

```
select(cars,matches('.a'))
```

16.arrange the data in ascending order of mpg

```
arrange(cars,mpg)
```

```
cars%>% group_by(cyl)%>% summarise(mx_mpg=max)
```

17.reading from external file

```
loan_data<-read.csv("example.csv")
```

18.print some aggregator for above csv file

```
csv_data <- read.csv(file ='sample.csv') new_csv <- subset(csv_data, department == "HR" & projects <10) print (new_csv)
```

19.editing a csv file

```
csv_data <- read.csv(file ='sample.csv') new_csv <- subset(csv_data, department == "HR" & projects <10) write.csv(new_csv, "new_sample.csv") new_data <-read.csv(file ='new_sample.csv') print(new_data)
```

20.Querying with csv file

```
csv_data <- read.csv(file ='sample.csv') min_pro <- min(csv_data$projects) print (min_pro)
```